

Understanding host-pathogen interactions and (co-)evolution through joint coalescent genealogies

Abstract for Jaques-Monod Conference *'Ecology and evolution of host-parasite interactions in a changing world'*, Roscoff 2026

Philip Wolper¹, Johannes Müller², and Aurélien Tellier¹

¹Chair of Population Genetics, Technical University of Munich, Germany

²Department of Mathematics, Technical University of Munich, Germany

September 8, 2026

1 Abstract

Host-pathogen coevolutionary systems are shaped by a complex interplay of epidemiological, ecological and evolutionary processes. Disease transmission, recovery and population dynamics of hosts and pathogens interact with mutation, selection, recombination and genetic drift to generate feedbacks that operate across multiple timescales and affect the resulting pattern of inheritance. Coevolutionary interactions involving reciprocal, frequency-dependent selection and fitness costs are affected by epidemiological feedback and ecological time scales. A large body of theoretical work exists on coevolution, typically focusing on solutions and equilibrium in continuous or discrete models of coevolution, successfully describing many complex models (Buckingham and Ashby 2022). It is however less understood, how coevolutionary interactions specifically shape the respective host and pathogens ancestral genealogical trees along their genomes (ancestral recombination graphs, ARGs) in the context of epidemiological spread and infection dynamics. While phylodynamics provides recent generalizations of epidemiological dynamics as coalescent processes (Didelot et al. 2025), similar coalescent-based considerations of coevolutionary models have been limited so far (with exception of Welch et al. 2005), despite recent advances simulating ARGs and tree sequences (Jeffery et al. 2026).

In this work we derive coalescence properties for of a pathogen embedded through vertical and horizontal transmission in a host genealogical process. We develop a semi-discrete stochastic model of host and pathogens with discrete, non-overlapping host reproduction and vertical and horizontal transmission of a pathogen between and within host generations, respectively. This model represents a pathogen, transmitted between generations and developing as epidemic cycles within host generations, such as the case for seasonal plant diseases and other semi-discrete host-pathogen systems. Our model builds upon the Wright-Fisher model, except with an embedded time-continuous infection model, and thus remains amenable to much established population genetics theory, including models of frequency-dependent selection (Tellier et al. 2021). Using deterministic approximations of the stochastic host and pathogen processes we derive pairwise coalescence probabilities for pathogens embedded in the host coalescence tree through vertical transmission. We show how pathogen coalescence is jointly affected by host coalescence and the epidemic process, dependent on the rate of vertical transmission c_v , the epidemiological contact rate β and the length of the epidemic outbreak G . To study stochastic effects and properties of joint host-pathogen ancestral trees (co-ARGs) for more than two samples, we implement this model using a two-species model in SLiM 5 (Haller and Messer 2023), using tree-sequence recording to track the ARGs of hosts and pathogens, allowing us to model the joint genealogies of hosts and pathogens explicitly forward in-time.

In future work, we wish to investigate the effects different epidemiological dynamics and transmission modes (eg. SIS, super-spreading) have on the joint properties of the host and pathogen genealogies. Additionally, we aim to extend our framework to more complex models of coevolution, including frequency-dependent selection and fitness costs within multiple types of hosts and pathogens, allowing for genetic interaction matrices common to coevolution. By investigating the behaviour of these joint genealogies under more complex coevolutionary and epidemiological models, we anticipate an improved understanding of the genomic signatures in host-pathogen interactions (Ebert and Fields 2020), better simulation-based inference methods (Märkle et al. 2024). We also discuss statistical methods of cophylogenetic host-pathogen species tree associations and how they could be applied to coevolutionary genealogies. Generally, we aim to provide a perspective of coevolution, inspired by existing tree-based approaches (such as phylodynamics, Plessis and Stadler 2015 or co-pylogenetics, Dismukes et al. 2022) and coalescent theory to explain joint host and pathogen genealogical processes. We hope, that formalizing coevolutionary interactions in the context joint genealogies could help connect such different tree-based methods across scales of host-pathogen interactions, including epidemiology, coevolution and speciation.

References

- Buckingham, Lydia J. and Ben Ashby (2022). “Coevolutionary Theory of Hosts and Parasites”. In: *Journal of Evolutionary Biology* 35.2, pp. 205–224. DOI: [10.1111/jeb.13981](https://doi.org/10.1111/jeb.13981). URL: <http://dx.doi.org/10.1111/jeb.13981>.
- Didelot, Xavier, David Helekal, and Ian Roberts (2025). “Ancestral Process for Infectious Disease Outbreaks With Superspreading”. In: *Journal of Theoretical Biology* 607.nil, p. 112109. DOI: [10.1016/j.jtbi.2025.112109](https://doi.org/10.1016/j.jtbi.2025.112109). URL: <http://dx.doi.org/10.1016/j.jtbi.2025.112109>.
- Dismukes, Wade et al. (2022). “Cophylogenetic Methods To Untangle the Evolutionary History of Ecological Interactions”. In: *Annual Review of Ecology, Evolution, and Systematics* 53. Volume 53, 2022, pp. 275–298. ISSN: 1545-2069. DOI: [10.1146/annurev-ecolsys-102320-112823](https://doi.org/10.1146/annurev-ecolsys-102320-112823). URL: <https://www.annualreviews.org/content/journals/10.1146/annurev-ecolsys-102320-112823>.
- Ebert, Dieter and Peter D Fields (2020). “Host–parasite Co-Evolution and Its Genomic Signature”. In: *Nature Reviews Genetics* 21.12, pp. 754–768.
- Haller, Benjamin C. and Philipp W. Messer (May 2023). “Slim 4: Multispecies Eco-Evolutionary Modeling”. In: *The American Naturalist* 201.5, E127–E139. ISSN: 1537-5323. DOI: [10.1086/723601](https://doi.org/10.1086/723601). URL: <http://dx.doi.org/10.1086/723601>.
- Jeffery, Ben et al. (2026). “Population-scale Ancestral Recombination Graphs with tskit 1.0”. In: *bioRxiv*. eprint: [2602.09649](https://doi.org/10.1101/2602.09649). URL: <https://arxiv.org/abs/2602.09649>.
- Märkle, Hanna et al. (2024). “Inference of Host-Pathogen Interaction Matrices From Genome-Wide Polymorphism Data”. In: *Molecular Biology and Evolution* 41.9, nil. DOI: [10.1093/molbev/msae176](https://doi.org/10.1093/molbev/msae176). URL: <http://dx.doi.org/10.1093/molbev/msae176>.
- Plessis, Louis du and Tanja Stadler (2015). “Getting To the Root of Epidemic Spread With Phylodynamic Analysis of Genomic Data”. In: *Trends in Microbiology* 23.7, pp. 383–386. DOI: [10.1016/j.tim.2015.04.007](https://doi.org/10.1016/j.tim.2015.04.007). URL: <http://dx.doi.org/10.1016/j.tim.2015.04.007>.
- Tellier, Aurélien et al. (2021). “Theory of Host-Parasite Coevolution: From Ecology to Genomics”. In: pp. 1–10. DOI: [10.1002/9780470015902.a0029063](https://doi.org/10.1002/9780470015902.a0029063). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470015902.a0029063>.
- Welch, David et al. (2005). “Integrating Genealogy and Epidemiology: the Ancestral Infection and Selection Graph As a Model for Reconstructing Host Virus Histories”. In: *Theoretical Population Biology* 68.1, pp. 65–75. DOI: [10.1016/j.tpb.2005.03.003](https://doi.org/10.1016/j.tpb.2005.03.003). URL: <http://dx.doi.org/10.1016/j.tpb.2005.03.003>.